Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mia rides 1/2 mile in 1/4 hour. What is her unit rate in miles per hour? Student's answer: 1/8

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3/4 pound of cheese costs $6. What is the price per pound? Student's answer: $4.50

Divide the cost by the weight.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mario walks 1 1/2 miles in 1/3 hour. What is his speed in miles per hour? Student's answer: 1/2 mile per hour

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Bella types 150 words in 2 1/2 minutes. How many words per minute? Student's answer: 375

Divide the words by the minutes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A snail moves 3/5 foot in 1/4 minute. How many feet per minute? Student's answer: 3/20

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is faster: 4 miles in 1/2 hour or 7 miles in 1 hour? Student says: 7 miles in 1 hour

Find the speed of each.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

7

Find the Error

7.RP.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is this table proportional? x = 1, 2, 3, 4 and y = 3, 6, 9, 10. Student says: Yes

Check whether y/x is the same for every pair.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

8

Find the Error

7.RP.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4 items cost $12 in total. What is the constant of proportionality for cost per item? Student's answer: 4

Divide the cost by the number of items.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The equation is y = 2.5x. What is y when x = 8? Student's answer: 10.5

Substitute 8 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 4x + 1$ a proportional relationship? Student says: Yes

Check whether the line passes through (0, 0).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table shows (2, 5) and (4, 10). What is the constant of proportionality? Student's answer: 2

Divide y by x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Sales commission is 10% of sales. If sales are $250, what is the commission? Student's answer: $2.50

Write 10% as a decimal and multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A restaurant bill is $40. What is a 15% tip? Student's answer: $15

Find 15% of $40.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An $80 item increases in price by 25%. What is the new price? Student's answer: $105

Find 25% of $80, then add it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A $25 book has an 8% sales tax. What is the total price? Student's answer: $2

Find the tax, then add it to the price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

18 of 24 students passed a test. What percent passed? Student's answer: 25%

Write the fraction and change it to a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A price drops from $60 to $45. What is the percent decrease? Student's answer: 15%

Find the change, then divide by the original price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A $200 item is marked 5% off. What is the sale price? Student's answer: $10

Find the discount, then subtract it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add -3 + 8. Student's answer: -11

Find the difference of the absolute values, then use the sign of the larger one.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 4 - (-6). Student's answer: -2

Subtracting a negative means adding its opposite.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add -2.5 + (-4). Student's answer: 6.5

Add the absolute values and keep the sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 3/4 + (-1/2). Student's answer: -1/4

Use a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract -7 - (-3). Student's answer: -10

Subtracting a negative means adding its opposite.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The temperature is −8°F and rises 13 degrees. What is the new temperature? Student's answer: −21°F

Start at −8 and move up 13.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply (-4) × (-6). Student's answer: -24

Use the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide (-12) ÷ 3. Student's answer: 4

Use the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(-2/3) \times (3/4)$. Student's answer: 1/2

Multiply the numerators and denominators.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide $-15 \div (-5/2)$. Student's answer: -6

Multiply by the reciprocal, then apply the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $0.5 \times (-0.4)$. Student's answer: 0.2

Multiply the numbers, then check the sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide (-3/4) ÷ (1/2). Student's answer: -3/8

Multiply by the reciprocal of 1/2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A team gains 3 points, then loses 5 points. What is the net change? Student's answer: 2

Write the gain as positive and the loss as negative.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The temperature falls from -4°F to -12°F. What is the change in temperature? Student's answer: 8°F

Subtract the starting value from the ending value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fish sinks 2.5 feet every minute for 4 minutes. What is its position change in feet? Student's answer: 10 feet

Multiply the rate by the time and use a negative sign for sinking.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

count gets $45 in deposits and $62.50 in withdrawals. What is the change in the balance? Student's answe

Write the withdrawals as negative amounts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

liver starts at -30 feet, rises 12 feet, then descends 20 feet. What is the final position? Student's answer: -18 f

Follow each move in order.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

arns $1.25 per item for 40 items and has costs of $60. Is it a profit or loss, and how much? Student's answer:

Write the earnings as positive and the costs as negative.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $3(x + 2) - x$. Student writes: $2x + 2$

Distribute, then combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 2x + 5 - 4x + 1. Student writes: -2x + 4

Combine like terms, keeping the signs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $4(2y - 1) + 3$. Student writes: $8y + 3$

Distribute the 4 to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify (1/2)x + (1/2)x. Student writes: x/2

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify -(3a - 2). Student writes: -3a - 2

Distribute the negative sign to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 0.5(4m) - 1.5m. Student writes: 3.5m

Multiply first, then combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write $4x + 10$ in factored form. Student writes: $4(x + 2.5)$

Find the greatest common factor of the coefficients.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An item is 15% off a price p. Which expression shows the sale price? Student writes: 0.15p

Think about what part is paid.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor 9y - 6. Student writes: 3(3y + 2)

Factor out the greatest common factor and check the signs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.4x + 0.6x equal to x? Student says: No

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tim says $5(2 + x) = 10 + x$. Is he correct? Student says: Yes

Distribute the 5 to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 12m + 18 in factored form. Student writes: 3(4m + 6)

Find the greatest common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Estimate 4.8 × 6.2. Student's estimate: 60

Round each number to a nearby whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Four notebooks cost $6.75 each, plus $3 for tax. What is the total? Student's answer: $24

Multiply, then add the tax.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2x + 3 = 11$. Student's answer: 8

Undo the addition, then undo the multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shirt costs $18 and the tax rate is 6%. What is the total price? Student's answer: $18.08

Find 6% of $18 and add it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 0.5x - 2 = 5. Student's answer: 7

Undo the subtraction, then the multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x - 2) = 12$. Student's answer: 2

Divide both sides by 3, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4x + 5 = 25$. Student's answer: 7.5

Subtract 5 from both sides, then divide by 4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x + 1) = 18$. Student's answer: 6

Divide by 3, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 2x - 7 = 9. Student's answer: 16

Add 7 to both sides, then divide by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $2x < 10$. Student writes: $x > 5$

Divide both sides by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $-3x \geq 12$. Student writes: $x \geq -4$

Divide both sides by -3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $x/2 + 1 > 4$. Student writes: $x > 3$

Undo the addition, then the division.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

p scale is 1 cm = 5 km. Two cities are 3.5 cm apart on the map. How far apart are they? Student's answer: 3

Multiply the map distance by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ving has a scale of 1 inch = 4 feet. A room is 12 feet long. How long is it in the drawing? Student's answer: 48

Divide the actual length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A scale drawing uses 1 cm = 2 m. A wall is 6 m long. How long is it in the drawing? Student's answer: 12 cm

Divide the actual length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

nap scale is 1 inch = 10 miles. Two towns are 2.5 inches apart. How far apart are they? Student's answer: 4 m

Multiply the map distance by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A photo that is 4 inches wide is enlarged to 10 inches wide. What is the scale factor? Student's answer: 0.4

Divide the new width by the original width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s a scale of 1 inch = 8 feet. A line on the drawing is 2.25 inches long. What is the actual length? Student's ans

Multiply the drawing length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 3 cm, 4 cm, and 5 cm? Student says: No

Check the triangle inequality.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 2 cm, 3 cm, and 6 cm? Student says: Yes

Check whether the two shorter sides add up to more than the longest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 60º, 60º, and 60º? Student says: No

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 90°, 50°, and 50°? Student says: Yes

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 40º, 60º, and 80º? Student says: No

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 1 cm, 2 cm, and 3 cm? Student says: Yes

Check whether the two shorter sides add up to more than the longest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lane cuts a cube parallel to one of its faces. What is the shape of the cross-section? Student's answer: triar

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lane cuts a cylinder parallel to its bases. What is the shape of the cross-section? Student's answer: rectan

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ne cuts through two opposite edges of a cube. What is the shape of the cross-section? Student's answer: tr

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What shapes are in the net of a square pyramid? Student's answer: 4 squares and 1 triangle

List each face.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many edges does a rectangular prism have? Student's answer: 8

Count the edges of the prism.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s through the tip of a cone and cuts straight down through its center. What is the cross-section? Student's

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the circumference of a circle with radius 3 cm. Student's answer: 9π cm

Use C = 2πr.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with diameter 10 cm. Student's answer: 100π cm²

Find the radius first, then use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with radius 5 cm. Student's answer: 10π cm²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A circle has a circumference of 12π cm. What is its diameter? Student's answer: 6 cm

Use C = πd.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with radius 2 cm. Student's answer: 2π cm²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a half-circle with a diameter of 8 cm. Student's answer: 16π cm²

Find the radius, find the full area, then take half.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are supplementary. One is 65°. What is the other? Student's answer: 25°

Supplementary angles add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are complementary. One is 35°. What is the other? Student's answer: 145°

Complementary angles add to 90°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are vertical angles. One is 70°. What is the other? Student's answer: 110°

Vertical angles are equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles of 45° and 60°. What is the third angle? Student's answer: 85°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are complementary. One is x and the other is 40°. Student says x = 140°

Complementary angles add to 90°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ngle of a triangle is 110°. One remote interior angle is 50°. What is the other remote interior angle? Student's

An exterior angle equals the sum of the two remote interior angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 8 cm and a height of 5 cm. What is its area? Student's answer: 40 cm²

Use A = 1/2 bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A circular table has a radius of 3 feet. What is its area, to the nearest hundredth? Student's answer: 18.85 ft²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

m has a base area of 12 square units and a height of 5 units. What is its volume? Student's answer: 17 cubic

Use V = Bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

r prism has a base area of 6 square units and a length of 10 units. What is its volume? Student's answer: 16

Use V = Bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box has a volume of 96 cubic cm. Its base is 8 cm by 4 cm. What is its height? Student's answer: 4 cm

Find the base area, then divide the volume by it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylinder has a radius of 2 cm and a height of 5 cm. What is its volume? Student's answer: 10π cm³

Use $V = \pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ers on the basketball team their favorite sport to learn about the whole school. Is this sample representativ

Think about who is in the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

om sample of 50 students is taken from a school of 500. Is the sample likely to be representative? Student s

Think about how the sample was chosen.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

first 15 people in line at an ice cream shop about flavors for all the students. Is this sample representative?

Think about who is in the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

100

Find the Error

7.SP.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What makes a sample representative of a population? Student's answer: It is a large sample.

Think about how the sample is chosen.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

101

Find the Error

7.SP.1

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tudents from a school of 400 found 10 prefer blue. Estimate how many of all 400 students prefer blue. Stud

Find the proportion in the sample, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eople from a town of 1,200 shows 15 people like pizza best. Estimate how many people in the town like pizza

Find the proportion in the sample, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ts from a school of 800, 20 students said they like the new lunch menu. Estimate how many students in the

Find the proportion, then scale up to the population.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e of 100 students, 60 said yes to a question. Estimate the number of yes answers in a population of 1,500. St

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

105

Find the Error

7.SP.2

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

of 25 students has 8 who like soccer. What is the estimated percent of the population who like soccer? Stu

Write the proportion as a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

200 from a city of 2,000 found 40 with brown eyes. Estimate the number of brown-eyed people in the city.

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

mple shows 60% prefer option A. Can we say exactly 60% of the population prefers option A? Student says:

Think about the variability in samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

of 80 students from a school of 800 shows 24 read every day. Estimate the number who read every day. St

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

up A has data 10, 12, 14. Group B has data 11, 13, 15. Which group has the higher mean? Student's answer: Grou

Find the mean of each group.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

up A has a range of 3 and Group B has a range of 10. Which group is more variable? Student's answer: Grou

Compare the spreads.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

as a median of 22 and Group B has a median of 33. What is the difference between the medians? Student's

Subtract the medians.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

hean 5 and group B has the mean 9. Both groups have a range of 2. Is the difference in means meaningful

Compare the difference in means to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Set 1 is 5, 5, 5. Set 2 is 1, 5, 9. Which set has more spread? Student's answer: Set 1

Compare how far values are from each other.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

mean of 50. Group A has a range of 5 and group B has a range of 25. Which group is less consistent? Stude

Compare the ranges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

deviation of 2. Group Y has a mean of 15 and a mean absolute deviation of 2. Is the difference in means lar

Compare the difference of means to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ool A has a mean of 7 hours of sleep. A random sample from School B has a mean of 7 hours. Student says:

Compare the means of the two samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ndom samples have medians of 40 and 42, with spread of 15 in each. Student says: The difference is mear

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

test score of 78 and class 2 has an average of 80. The spread in each class is about 20. Student says: Class

Think about the spread and the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

as a mean of 30 and a spread of 4. Group B has a mean of 36 and a spread of 4. Student says: The groups

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

nean of 100 with a spread of 50. A sample from population 2 has a mean of 110 with a spread of 50. Student

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A weather app says the chance of rain is 1.5. Can this be correct? Student says: Yes

Think about the range of probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of an impossible event? Student's answer: 1

Think about the range of probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of rolling a number less than 7 on a standard die? Student's answer: 0

Think about whether the event always happens.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of drawing a red card from a deck with 26 red cards out of 52? Student's answer: 2

Write the probability as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Letters A, B, C, and D are placed in a bag. What is the probability of picking a vowel? Student's answer: 1/2

Count the vowels out of the total letters.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is most likely: an event with probability 0.2, 0.7, or 0.45? Student's answer: 0.2

Compare the probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s flipped 50 times and lands heads 28 times. What is the experimental probability of heads? Student's answ

Divide heads by total flips.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

her is spun 100 times and lands on red 20 times. What is the experimental probability of red? Student's answ

Divide red outcomes by total spins.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

heoretical probability of heads of 0.5. A student flips it 10 times and gets 3 heads. Student says: The coin mu

Think about small samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s rolled 60 times, and 6 comes up 12 times. What is the experimental probability of rolling a 6? Student's ans

Divide the number of 6s by the total rolls.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

is tossed 10 times and lands heads 3 times. What is the experimental probability of heads? Student's answ

Divide heads by total tosses.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

etball player makes 7 of 10 free throws. Estimate the probability she makes her next free throw. Student's an

Divide made shots by total shots.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fair six-sided die is rolled. What is the probability of rolling a 4? Student's answer: 1/3

Assume each outcome is equally likely.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

l gives outcomes A, B, C, and D with probabilities 0.2, 0.3, and 0.4 for A, B, and C. What is P(D)? Student's ans

The probabilities must add to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A spinner has 8 equal sections, and 3 are blue. What is P(blue)? Student's answer: 3/5

Count the blue sections out of all sections.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bag has 12 marbles, 4 of them red. A marble is picked at random. What is P(red)? Student's answer: 4/8

Count the red marbles out of all marbles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

inner has 4 equal sections with probabilities 0.2, 0.2, 0.2, and 0.2. Student says: This is a valid probability mo

Check whether the probabilities add to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s chosen at random from the 26 letters of the alphabet. There are 5 vowels. What is P(vowel)? Student's ans

Count the vowels out of all letters.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two fair coins are flipped. What is the probability of getting two heads? Student's answer: 1/2

List all possible outcomes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two fair dice are rolled. What is the probability that the sum is 7? Student's answer: 1/12

List the outcomes that make 7.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A coin is flipped and a die is rolled. How many total outcomes are there? Student's answer: 8

Multiply the number of outcomes for each event.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Original task. Format inspired by error analysis routines.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 red and 2 blue marbles. Two marbles are picked without replacement. What is P(both red)? Student's an

Use the marbles left after the first pick.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A student has 2 shirts and 3 pairs of pants. How many different outfits can she make? Student's answer: 5

Multiply the number of choices.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two coins are flipped. What is the probability of getting at least one head? Student's answer: 1/2

Use the outcomes or the complement.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.